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A Robust Preprocessing–EfficientNet-B4 Pipeline for Multi-Disease Retinal Fundus Image Classification

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***Abstract*—Automated analysis of retinal fundus images can help in the early screening of vision-threatening diseases. This paper introduces a Deep learning method of four-class retinal disease classification based on Transfer learning of EfficientNet-B4. A high-quality dataset of 3588 retinal images was used and labeled as Age-related Macular Degeneration (409), Glaucoma (1007), Diabetic Retinopathy (1098), and Normal (1074). Preprocessing of images involved extracting the retinal field-of-view, center square cropping, re-sizing at 380*380, CLAHE in LAB color space to improve visibility of lesions and normalize intensities on a [0,1] scale. The dataset was split into training, validation, and test datasets, and the model was trained for 25 epochs by the frozen-backbone stage and fine-tuning with class weighting to handle imbalance. Performance was assessed with the help of accuracy, per-class Precision/Recall/F1-score, and confusion matrix. The proposed model showed 90.52% test accuracy with good F1-scores between classes and low inter-class confusion, showing the use of EfficientNet-B4 as a good baseline for automated analysis of retinal diseases.**

Keywords—Retinal fundus images, EfficientNet-B4, Deep learning, Transfer learning, Diabetic retinopathy, Glaucoma detection

I. INTRODUCTION

Vision-threatening eye diseases are a major global health problem, where delay in diagnosis can cause irreversible impairment and a high socioeconomic burden. Modern ophthalmology faces a broad spectrum of diseases too - from infectious causes still emerging and re-emerging in the 21st Century [1], the systemic vascular diseases like hypertensive eye disease which can develop silently while affecting ocular tissues [2], and chronic surface pathologies like dry eye disease where gaps in care and treatment pathways are still reported despite a growing armamentarium of clinical options [3]. Parallel advances in therapy, such as nanozyme-based eye drops designed to mitigate oxidative and inflammatory mechanisms in dry eye disease [4], further highlight the importance of early detection and an effective triage mechanism so that early interventions can be instituted and effectively monitored. However, population-scale screening is often limited by a lack of available specialists, inter-observer variability, and the time-consuming nature of manual image interpretation. Retinal fundus photography offers a non-invasive view into the anatomy of the posterior segment, thus allowing the visualization of vascular changes, optic nerve head changes, and lesion patterns important to common retinal pathologies. In particular, Age-related Macular Degeneration (AMD), Glaucoma, and Diabetic Retinopathy (DR) are some of the top causes of progressive vision loss and need to be referred and followed up promptly. Automated fundus-image analysis has therefore become an important field in research where deep learning (DL) models can learn discriminative representations directly from pixels and can facilitate computer-aided workflows for screening. Recent work has shown the benefit of convolutional neural networks (CNNs) in the detection and classification of DR [5] and also multi-class eye disease recognition with transfer learning using EfficientNet-family backbones to achieve better accuracy and efficiency [6]. Alternative deep architectures, such as InceptionResNetV2-based pipelines, have also been explored for feature extraction and robustness under real-world imaging variability [7]. Additionally, ongoing DL-focused research continues to emphasize the improvements in lesion sensitivity and automated DR classification performance, further supporting the feasibility of deploying learning-based systems in screening contexts [8]. Despite this progress, effective multi-disease fundus classification is difficult because of heterogeneous acquisition conditions (illumination variations, contrast differences, and field-of-view, etc.), subtle inter-class visual similarities, and class imbalance leading to biased predictions towards majority classes. Motivated by the above limitations, this paper proposes a robust preprocessing

- EfficientNet-B4 pipeline for retinal fundus image classification of four classes (AMD, Glaucoma, DR, and Normal). By standardizing the retinal FOV, lesion-preserving contrast enhancement, and the transfer learning approach with staged fine-tuning and imbalance-aware optimization, the proposed approach is aimed at providing a powerful and reproducible baseline that can be used for screening-oriented applications. The main contributions are threefold: (i) a consistent preprocessing workflow customized to fundus imagery, (ii) an EfficientNet-B4 transfer learning approach with training of frozen backbone and fine-tuning, and (iii) extensive evaluation with per-class evaluation and confusion matrix analysis to understand clinically relevant error patterns across the disease categories.

II. Literature Review

Recent advances in artificial intelligence (AI) and deep learning (DL) have had a major impact on automated ophthalmic image analysis for screening of retinal diseases in particular. Wahab Sait et al. [9] introduced an AI-based eye disease classification model, which demonstrates the increasing feasibility of data-driven pipelines in ocular diagnosis. Their work strengthens a key trend in the field, which is to harness learned visual representations to decrease reliance on handcrafted clinical rules and to enable scalable screening. Complementing this direction, Bali and Mansotra et al. [10] reported a systematic review of DL techniques for eye disease prediction, and conclude that transfer learning and modern CNN architectures consistently outperform traditional machine learning pipelines when trained on adequately curated datasets. However, the review also highlighted the persistent limitations, including dataset bias, inconsistent preprocessing, and the absence of cross-dataset validation, reducing the real-world generalizability. A major part of the literature is devoted to the imaging of the retinal fundus for the detection of diabetic retinopathy (DR), glaucoma, and other pathologies. Jain and Rathour et al. [11] investigated DR detection with InceptionV3 and showed that deep pretrained backbones are able to extract the discriminative lesion features such as microaneurysms and hemorrhages. Similarly, Jain and Rathour and others [5], [8] explored CNN-based DR detection and classification strategies, further validating the power of end-to-end feature learning for grading the disease and multi-class DR recognition. Albelaihi and Ibrahim et al. [12] presented "DeepDiabetic," a diabetic eye disease identification system that relies on deep learning neural networks, which reflects a growing trend toward consolidated systems that are able to discriminate multiple diabetic eye diseases, instead of binary discrimination of DR/non-DR eye diseases. These studies collectively demonstrate that high-capacity CNNs can yield strong performance, but they also demonstrate that performance can be sensitive to things such as preprocessing, lesion visibility, and class imbalance - factors which are often very different across data sets. Beyond DR, multi-disease classification has received impetus as a more clinical, realistic objective. Shamsan et al. [13] suggested automatic color fundus image classification based on hybrid features, i.e., a combination of engineered features and learning-based features to enhance the discrimination among types of diseases. While hybrid approaches can yield interpretability and robustness in smaller datasets, they can be underutilizing the representational power of modern end-to-end CNNs in scenarios where larger annotated corpora are available. Muthukannan et al. [14] presented an optimized CNN framework for the detection of multiple eye diseases, which shows the ongoing effort to customize network design and optimization for the nature of ophthalmic data. In parallel, Kaushik and Sharma et al. [6], [15] explored frameworks based on EfficientNetB3 for multi-class eye disease classification that employs the importance of compound-scaled architectures for achieving improved accuracy-efficiency tradeoffs than earlier CNNs. Their study findings have shown that the EfficientNet variants can be useful for multi-class ocular diagnosis, especially when combined with transfer learning and task-specific fine-tuning. While the imaging modality of the fundus is widely used, a new complementary modality for the diagnosis of structural abnormalities of the retina has emerged: optical coherence tomography (OCT). Puneet et al. [16] and Kaushik and Choudhary et al. [17], [18] proposed an OCT-based framework for eye disease detection using deep CNNs (including VGG16-based modeling), demonstrating that the cross-sectional retinal layer patterns can be highly discriminative for the detection of eye diseases. These works suggest the modality choice has a strong effect on feature learning: fundus images focus on the superficial appearance of lesions and vascular morphology, while OCT provides depth-resolved anatomic changes. Despite these benefits, OCT pipelines are often plagued with issues that involve variability in scan quality, device differences, and domain shift between clinical sites, which again stresses the importance of robust preprocessing and evaluation strategies. A further trend in the literature is the growing application of transfer learning to various medical imaging challenges, which is in favor of the motivation of EfficientNet-based approaches. Kaushik and Sharma et al. [19] employed transfer learning for cataract detection in fundus images and showed the feasibility of the use of pretrained backbones for ophthalmic diagnosis other than retina disorders. Additionally, EfficientNet family-based modeling has been successfully extended to non-ocular medical classification tasks, for example, diabetic foot ulcer classification [20], classifying skin diseases with EfficientNetB5 [21], and detecting leukemia with EfficientNetB3 [22]. Although these studies refer to different areas of clinical practice, they collectively support the more general claim that EfficientNet-family networks are generalizable by transfer learning, especially when preprocessing is used to standardize the distribution of inputs and training strategies are used to mitigate class imbalance. In general, previous studies prove that DL methods, in particular, transfer learning using modern CNN backbones, could provide powerful performance in eye disease detection for modality and task diversity [9] - [22]. Nevertheless, key gaps remain: inconsistent or under-specified preprocessing pipelines, limiting reproducibility, insufficient management of class imbalance in multi-disease settings, and insufficient reporting of clinically meaningful confusion patterns and per-class reliability. These limitations provide a motivation for the current work that focuses on a powerful and well-defined preprocessing pipeline for fundus images, and an EfficientNet-based transfer learning approach tailored for four classes of retinal diseases and well-defined per-class evaluation.

III. METHODOLOGY

A. Data Collection

This study uses a high-quality retinal fundus image dataset curated for automated analysis of four clinical categories: Age-related Macular Degeneration (AMD), Glaucoma, Diabetic Retinopathy (DR), and Normal [23]. The dataset has a total of 3,588 images that consist of 409 AMD, 1,007 Glaucoma, 1,098 DR, and 1,074 Normal samples, as shown in Fig. 1. Retinal images are given in standard formats (e.g., JPG/PNG) and contain obvious visual manifestations of lesions and disease-related abnormalities of various sizes and shapes. A single label of disease categories is assigned to each image, and this enables supervised multi-class classification. The dataset is imbalanced in terms of class distribution, with the class of AMD being the minority class, and this is taken into account while training and evaluating the model to prevent biased predictions.

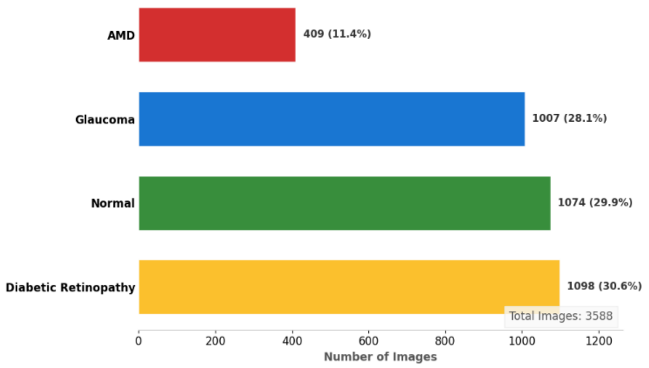

Fig. 1. Class-wise distribution of the knee osteoporosis dataset

B. Data Preprocessing

All images of the retinal fundus were preprocessed to have a consistent input quality for EfficientNet-B4. Each image was first converted to RGB and checked for readability. In order to eliminate the background effects, the retinal field-of-view (FOV) was obtained by thresholding the dark regions and selecting the largest connected component, followed by cropping with the FOV bounding box. The cropped image was then center-cropped to a square shape to retain content from the retina while allowing for equal-sized scaling. Next, resizing of images to 380 × 380 pixels was done as per EfficientNet-B4 input requirements. Contrast-Limited Adaptive Histogram Equalization (CLAHE) was used on the L-channel in the LAB color space to increase the vessel and lesion visibility in varying levels of illumination. Finally, pixel intensities were normalized to [0,1] by dividing by 255, giving a float tensor ready for training, as illustrated in Fig. 2.

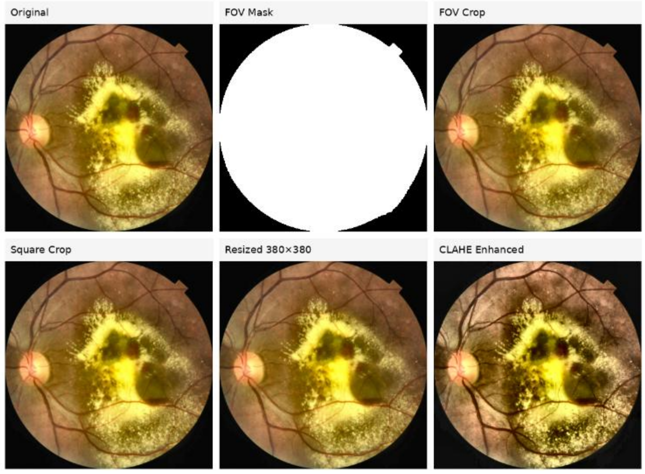

Fig. 2. Step-by-Step Retinal Image Preprocessing

The retinal data set (N = 3,588) was divided using a stratified split in order to maintain the same class proportions between subsets. A 70% / 15% / 15% partition was used for training, validation, and testing, respectively. This led to 2,512 training images (70.0%), 538 validation images (15.0%), and 538 test images (15.0%). Class-wise allocation was the following: AMD (409): 286/61/62 (train/val/test); Glaucoma (1,007): 705/151/151; Diabetic Retinopathy (1,098): 769/165/164; Normal (1,074): 752/161/161, as shown in Fig. 3. The validation set was utilized for model selection and hyperparameter tuning, and the test set was not seen until final reporting of performance to avoid biased evaluation.

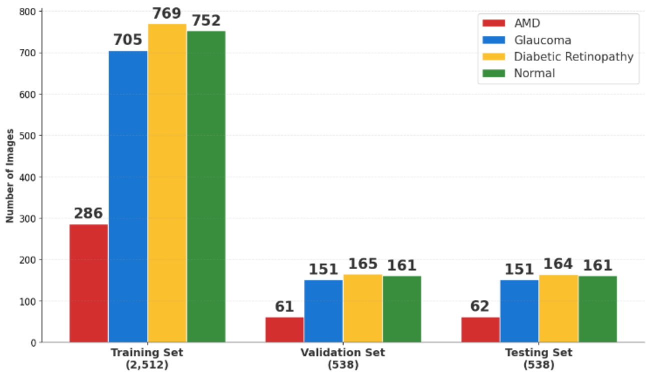

Fig. 3. Stratified Partitioning of Retinal Fundus Dataset

C. Model Selection and Architecture

EfficientNet-B4 was chosen as the backbone for multi-class retinal disease classification because of its excellent accuracy-efficiency tradeoff and compound scaling strategy (i.e., joint scaling of depth, width, and input resolution). The architecture of the proposed system is presented in Fig. 4. The model accepts the preprocessed images of the fundus of dimensions 380×380×3. EfficientNet-B4 includes a convolutional stem, after that, stacked MBConv blocks with depthwise separable convolutions and squeeze-and-excitation (SE) attention to facilitate channel-wise feature learning. These blocks capture clinically relevant patterns such as alterations in vessels, hemorrhages, exudates, and the optic discs very well. After the last convolution stage, global average pooling is applied to compress feature maps into a low-dimensional representation. A task-specific dense head was added with dropout regularization, and the last layer is Softmax (4 units) with the prediction for AMD, Glaucoma, DR, and Normal classes.

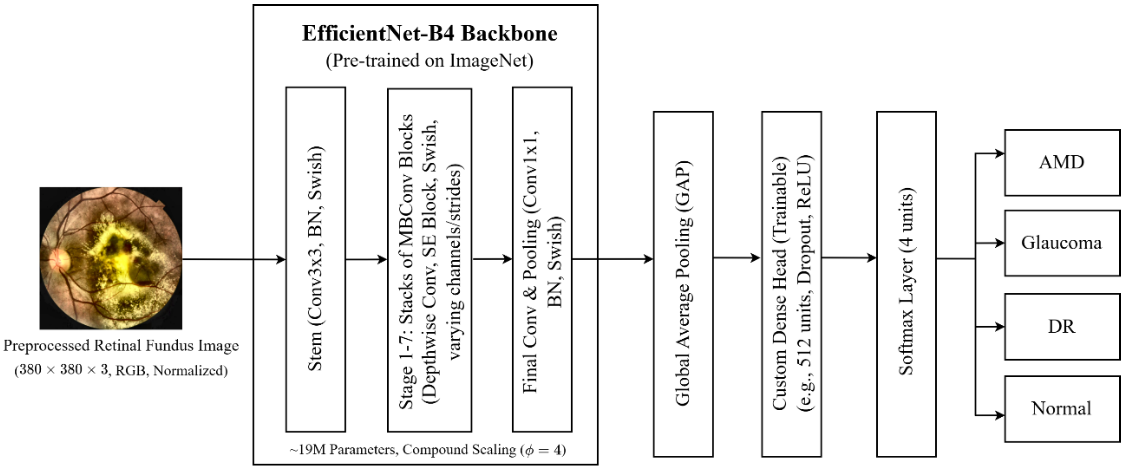

Fig. 4. EfficientNet-B4 pipeline for four-class retinal disease classification

D. Training Process and Evaluation

EfficientNet-B4 was trained with transfer learning on the preprocessed fundus images (380×380, normalized to [0,1]) with four-class AMD, Glaucoma, DR, and Normal retinal disease classification. The dataset was divided by stratification into train, validation, and test datasets to maintain the proportion of classes. Training was executed for 25 epochs in 2 phases: First phase training, where the ImageNet pretrained backbone was frozen and only the custom Softmax classification head was trained, and second phase fine-tuning, where the upper layers were unfrozen and optimized with a lower learning rate to learn subtle lesions pattern. Categorical cross-entropy loss was used with Adam/AdamW, and class weight was used to mitigate the effect of class weights, especially for AMD. Early stopping and best model checkpointing based on validation loss were used to limit overfitting. Final evaluation on the test set showed accuracy, precision per class, recall, F1-score, and a confusion matrix.

IV. RESULTS

A. Training and validation loss

Training loss reduced continuously over 25 epochs, and thus, this shows the effective convergence of the EfficientNet-B4 model. The loss dropped from 1.1500 (Epoch 1) to 0.5587 (Epoch 5), showing a quick adaptation to retinal disease features. Continued optimization further reduced training loss to 0.1973 by Epoch 12 and less than 0.100 after Epoch 20 to 0.0878 at Epoch 25. Validation loss also showed a similar decrease from 1.0217 (Epoch 1) to 0.5961 (Epoch 5) and 0.2884 (Epoch 12). The lowest validation loss came out to be 0.1552 (Epoch 25). As illustrated in Fig. 5, minor fluctuations seen between Epochs 18-23 reflect a mild degree of overfitting; however, the general downwards trend is evidence of stable learning and efficient generalization across categories of retinal disease.

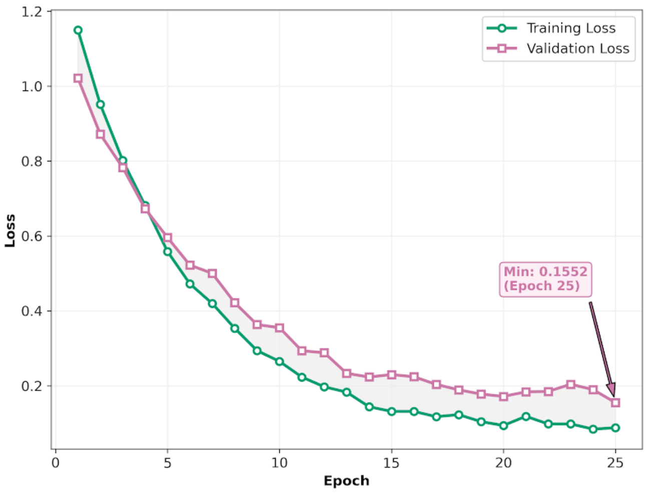

Fig. 5. Training and Validation Loss During Model Training

B. Training and Validation Accuracy

Training accuracy improved progressively from 0.5250 (Epoch 1) to 0.7635 (Epoch 5), indicating progressive feature learning in those early epochs. Accuracy exceeded 0.90 at Epoch 12 (0.9018) and exceeded 0.93 in later stages of accuracy, with a peak at 0.9500 (Epochs 22 and 24) followed by a slight decrease in accuracy at Epoch 25 (0.9334). Validation accuracy has also shown constant improvement, as depicted in Fig.6, going from 0.4691 (Epoch 1) to 0.7127 (Epoch 6) to 0.8486 (Epoch 12). The maximum validation accuracy of 0.8924 was recorded at epoch 17, followed by stabilization of the value around 0.87-0.89. The close match between training and validation accuracy indicates controlled overfitting and good generalization performance of multi-class retinal disease classification.

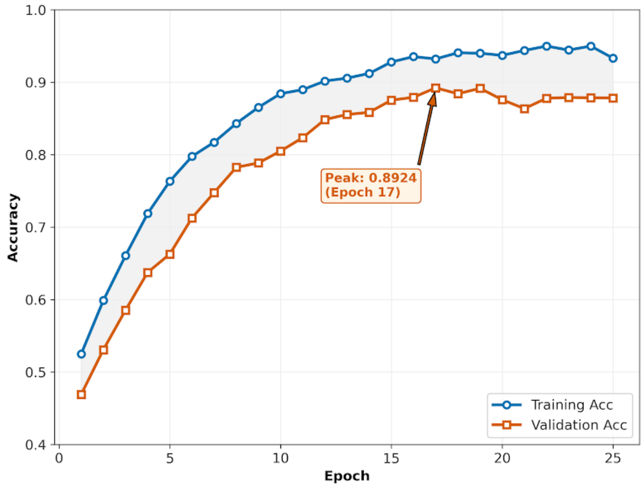

Fig. 6. Training and Validation Accuracy During Model Training

C. Confusion Matrix Analysis

The confusion matrix on the test set of 538 images, shown in Fig. 7, has a strong diagonal dominance in all 4 classes. AMD gave 56 correct predictions out of 62, while the errors were mostly located in Glaucoma (3), DR (2), and Normal (1). Glaucoma was at 135/151 correct, with the greatest confusion to DR (8) and lesser leakage to AMD (4) and Normal (4). DR obtained 149/164 correct, misclassified mostly as Glaucoma (6) and AMD (5), and rarely as Normal (4). Normal scored 147/161 correct, most mistakes were predicted as DR (7) and fewer as Glaucoma (4) or AMD (3). Overall, cross-class confusion is limited and mostly between clinically similar Glaucoma - DR patterns. This behavior is preferred in screening where misrouting across diseases is expensive.

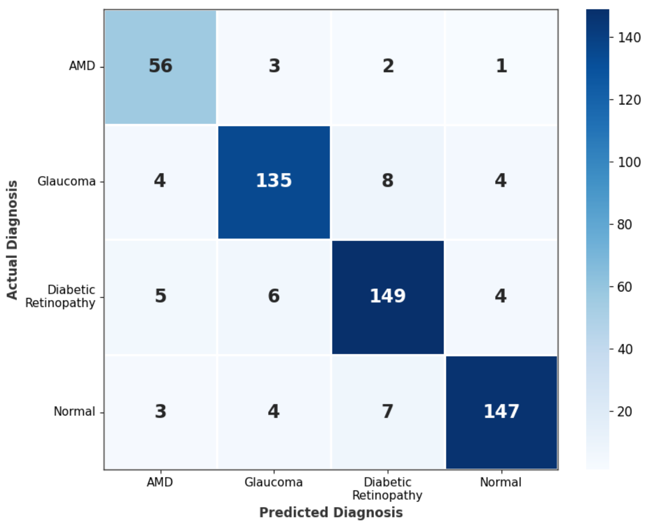

Fig. 7. Confusion Matrix Depicting Classification Performance

D. Classification Report Analysis

The classification report, summarized in Table I, shows consistent multi-class performance with an accuracy of 90.52%. AMD (support=62) had 82.35% precision and 90.32% recall (F1=86.15), which means that most of the AMD images were found, but some of the non-AMD images were also predicted to be AMD. Glaucoma (support=151) achieved 91.22% precision-recall 89.40% (F1=90.30). DR (support=164) gave a balanced precision/recall of 90.85% (F1=90.85), demonstrating stable detection of lesion-driven changes. Normal (support=161) had 93.04% precision and 91.30% recall (F1=92.16), which is low false alarms of disease. Macro averages (Precision=89.37, Recall=90.47, F1=89.87) as well as weighted averages (F1=90.55) show powerful results under class imbalance in practice.

TABLE I. Classification Report Analysis

Class	Precision %	Recall %	F1-Score %	Specificity %	Support %
AMD	82.35	90.32	86.15	97.48	62
Glaucoma	91.22	89.40	90.30	96.64	151
Diabetic Retinopathy	90.85	90.85	90.85	95.99	164
Normal	93.04	91.30	92.16	97.08	161
Overall Accuracy			90.52		538
Macro Average	89.37	90.47	89.87	96.80	538
Weighted Average	90.63	90.52	90.55	96.72	538

V. Conclusion

This work introduced a deep learning framework for automated classification of retinal disease based on ImageNet pre-trained EfficientNet-B4. A four-class data set of fundus images consisting of AMD, Glaucoma, Diabetic Retinopathy, and Normal images was preprocessed by Field-Of-View extraction, square centering, image resizing to 380x380, CLAHE-based contrast enhancement, and intensity normalization. The model has been trained using transfer learning and fine-tuning, where it converged stably in 25 epochs. On the test set (538 images), the proposed approach yielded 90.52% overall accuracy, with good class-wise F1-scores of Glaucoma (90.30), DR (90.85), Normal (92.16), and competitive performance for minority AMD (86.15). Confusion analysis revealed limited misclassification with most errors between clinically similar categories, such as Glaucoma and DR. Future work will involve focusing on minority classes (particularly AMD) with advanced class balancing, focal loss, and threshold calibration. Incorporating lesion-aware attention, multi-scale feature fusion, and explainability initiatives (e.g., Grad-CAM) can boost clinical trust and interpretability. Extending evaluation to cross-dataset tests, real-world clinical scenarios, and image quality variations, as well as deployment-focused compression (quantization/pruning), will further enable practical screening use cases.

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